

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

May 2012

EC204 Electronics Instruments & Measurements

Date: 04.05.2012, Friday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Give definitions of following terms with respect to electronics measurements: [05]

1. Accuracy
2. Error
3. Precision
4. Resolution
5. Probable error

(b) Calculate [06]

- a. Arithmetic mean
- b. The average deviation
- c. The standard deviation

For the following data

1. Measurements of resistances
147.5, 147.6, 147.4, 147.6, 147.5, 147.2
2. Measurements of Voltages
49.6, 49.3, 49.7, 50.1, 50.2
3. Measurements of frequency
19.2, 19.4, 19.3, 19.1, 19.2, 19.7, 19.3

(c) Explain why: [04]

1. Maxwell's bridge is not suitable for high Q-coils?
2. Maxwell's bridge is not suitable for low Q-coil?

Q - 2 (a) Describe an AC Schering bridge. How will you measure the following with such a bridge: [06]

1. unknown capacitance
2. leakage resistance of the capacitor

(b) Draw the basic block diagram of an oscilloscope and explain the function of each block. [04]

OR

Q - 2 (a) Describe the circuit of Kelvin double bridge used for measurement of low resistance. [06]

(b) Explain the working principle of a digital multimeter. How current and resistance can be measured in digital multimeter? [04]

Q - 3 (a) Why a delay line used in the vertical section of Oscilloscope? Explain the operation circuit of the delay line. [04]

- (b) The impedance of the basic AC bridge of Figure 1 are given as follows: [06]
- $Z_1 = 50 \Omega \angle 80^\circ$ (inductive impedance)
 $Z_2 = 125 \Omega$ (pure resistance)
 $Z_3 = 200 \Omega \angle 30^\circ$ (inductive impedance)
 $Z_4 = \text{Unknown}$
 Determine the constants of the unknown arm.

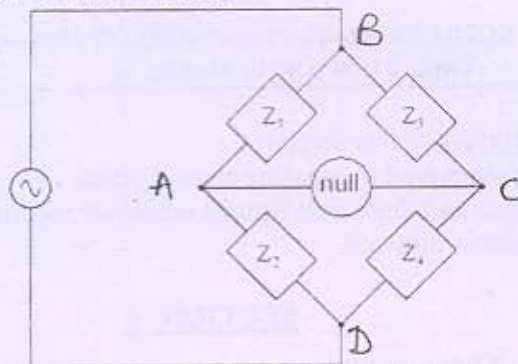


Figure 1
OR

- Q - 3 (a) The AC bridge of Figure 1 is in balance with the following constants: arm AB, $R=450\Omega$, [06]
 arm BC, $R=300\Omega$ in series with $C=0.265 \mu\text{F}$; arm CD, unknown; arm DA, $R=200\Omega$ in series with $L=15.9 \text{ mH}$. The oscillator frequency is 1 kHz. Find the constants of arm CD.
 (b) Derive an expression for the gage factor in case of strain gage. [04]

SECTION - II

- Q - 4 (a) Differentiate the following: [06]
1. Active Vs Passive transducers
 2. RTD Vs Thermocouple
 3. Lumped-Parameter Vs Distributed Parameter Delay line
- (b) An input signal was sampled for 4 seconds at a sampling rate of 20 kHz using a 10 bit [05]
 Conversion. Find resolution, total frequency display and dynamic range available from given input signal.
 (c) Draw a block diagram of fast Fourier Transform (FFT) spectrum analyzer. [04]
- Q - 5 (a) Differentiate Passive and Active matrix displays. Explain the basic concept of each with [06]
 advantages and disadvantages.
 (b) What are the advantages of dual trace over dual beam CROs for multiple trace? [04]

OR

- Q - 5 (a) Explain common cathode type 7-segment display with a logic diagram. [04]
 (b) Explain the operating principles of Liquid Crystal Display. [06]
- Q - 6 Write short notes on following (any two): [10]
- a. LVDT
 - b. Capacitive transducers
 - c. Digital Storage Oscilloscope (DSO)
 - d. Photo cells
